

Sound Over Spelling: A Comparative Study of Phonetic vs. Subword Tokenization for Hindi/Hinglish Language Modeling

Anushka Shakya, Anushka Rajput, Anushka Kaushal, Aditi Singh, Aditya Singh

Department of Computer Science and Engineering (Artificial Intelligence),

KIET Group of Institutions, India

Abstract

Spelling variation is a persistent characteristic of romanized Hindi-English code-mixed text (Hinglish), where the same word may occur in multiple written forms without a standardized orthographic convention. Standard subword tokenization operates directly on these spellings and may therefore encode surface-form variation as distinct token sequences. This study investigates whether phonetic tokenization, which represents text according to pronunciation, provides greater robustness to orthographic noise than conventional subword tokenization when language models are trained at the same scale. The study compares the two representations at both the tokenizer and language-model levels, including token compactness, spelling-variant collisions, language-model perplexity, and downstream sentiment classification under clean and spelling-perturbed conditions.

Index Terms: Hinglish, Hindi-English code-mixing, phonetic tokenization, BPE, subword tokenization, spelling variation, language modeling, robustness

I. INTRODUCTION

Tokenization determines how textual input is represented before it is processed by a language model. Contemporary language models predominantly rely on orthography-based subword schemes such as Byte Pair Encoding (BPE), which construct a vocabulary from recurring character sequences. While effective for standardized written languages, such representations can be sensitive to spelling variation. This issue is particularly relevant to Hinglish, in which Hindi is frequently written using the Roman alphabet and the absence of a fixed transliteration standard permits multiple spellings of the same underlying word.

Phonetic tokenization provides an alternative representation in which text is mapped from graphemes to pronunciation units before language-model training. Rather than treating spelling as the primary signal, this representation preserves information that is shared across orthographic variants with similar pronunciation. Prior work has demonstrated the feasibility of phoneme-based language modeling and has also

shown that conventional tokenization can obscure phonological structure. The present study extends these observations to Hinglish and focuses specifically on robustness to spelling variation.

The central research question is: Does phonetic tokenization make language models more robust to spelling errors and orthographic noise than standard subword tokenization, when trained at the same scale? To address this question, the study evaluates the two tokenization approaches under controlled conditions and separates tokenizer-level measurements from model-level evaluation.

II. RELATED WORK

A. Phoneme-Based Language Modeling

Goriely et al. (2024) provide a foundational proof of concept for training language models directly on phonemes rather than orthographic text. Working within the BabyLM challenge framework, they develop a pipeline that converts the 100-million-word BabyLM pre-training corpus and associated grammatical benchmarks into continuous streams of phonemes and subsequently train and evaluate a GPT-2-style model on this phonemic representation. Their results demonstrate a trade-off between phonemic and orthographic modeling: phoneme-based training leads to a small reduction in performance on conventional orthography-based language understanding benchmarks, while providing benefits on sound-based and speech-specific tasks. Their analysis also identifies utterance-boundary markers as important components of an effective phoneme-level training setup. Their experiments are conducted on clean, standardized English text. The present study extends this line of work to Hindi/Hinglish, where orthographic conventions are substantially less consistent and spelling variation is an inherent characteristic of romanized code-mixed text.

B. Tokenization as a Source of Phonological Information Loss

Liao and Shi (2026) examine the phonological consequences of standard subword tokenization and shift the discussion toward what information conventional tokenization may discard. Through experiments on text-only language models, they show that subword and BPE-style tokenization can weaken representations of phonological structure, including local properties such as rhyme and broader properties such as syllabification. They introduce the syllabification-tokenization alignment distance (STAD) to quantify divergence between learned token boundaries and natural syllable boundaries. Their findings further show that lightweight fine-tuning on an International Phonetic Alphabet (IPA)-based representation can recover a substantial portion of the lost phonological knowledge. This work motivates the present study's examination of whether phonetic representations can similarly mitigate problems caused by orthographic inconsistency in romanized Hindi-English text.

C. Phonetic Representations for Cross-Lingual Fairness

Miletić et al. (2026) investigate phonetic representations from a multilingual perspective, focusing on tokenization quality and fairness across languages and writing systems. They report that conventional multilingual subword tokenizers can favor high-resource languages, while byte-level approaches may disadvantage scripts with higher bytes-per-character ratios. They propose tokenizing International Phonetic Alphabet (IPA) representations rather than raw orthographic text. Across experiments covering 24 languages and 14 scripts, they find that IPA-based tokenization improves tokenization quality and generalizes more effectively to unseen languages and scripts, with notable benefits for non-Latin-script languages. Although this work addresses a multilingual setting rather than the single language-pair focus of the present study, it provides relevant evidence for the broader use of phonetic representations.

D. Positioning of the Present Study

Taken together, prior studies establish that phoneme-based language modeling is viable, that conventional subword tokenization can obscure aspects of phonological structure, and that phonetic or IPA-based representations can provide measurable benefits across languages and scripts. However, these studies do not directly examine naturally occurring spelling variation within a single code-mixed language community using Roman script. Hinglish provides a relevant test case because the same Hindi word may appear in multiple Romanized spellings without a standardized orthographic convention. The present study therefore focuses on whether representing text through pronunciation improves robustness to spelling variation compared with standard subword tokenization under comparable training conditions.

III. METHODOLOGY

A. Experimental Design

The study compares a BPE-based tokenizer with a phonetic tokenizer under controlled conditions. Both representations are evaluated using matched vocabulary settings where applicable. The tokenizer-level analysis measures vocabulary size, average tokens per sentence, and the ability to map spelling variants to shared representations. The model-level experiments use the same language-model configuration for both tokenization schemes so that observed differences can be attributed primarily to the input representation.

B. Datasets

The experimental pipeline is developed first on the IIT-H code-mixed corpus to enable rapid iteration and debugging. The analysis is then intended to scale to L3Cube-HingCorpus. A Hinglish-Hindi transliteration dataset is used to construct spelling-variant pairs for robustness analysis, while a labeled Hinglish sentiment corpus provides the downstream evaluation task. The spelling-variant evaluation is central to testing whether phonetic representations are less sensitive to orthographic variation.

C. Tokenization Pipeline

For the phonetic representation, Hinglish text is converted from grapheme sequences to phoneme sequences using a Hindi-capable grapheme-to-phoneme system. The resulting phoneme inventory is mapped to integer identifiers for language-model input. The baseline applies a standard BPE tokenizer to the same preprocessed text. Both tokenizers are evaluated before model training to measure compactness and spelling-variant collisions.

D. Robustness Evaluation

Robustness is evaluated by comparing model performance on clean and spelling-perturbed inputs. Spelling variants derived from the transliteration data provide a controlled source of orthographic noise. The downstream sentiment classifier is evaluated under both conditions, allowing the change in performance caused by spelling variation to be measured separately for the two tokenization schemes.

IV. EXPECTED EVALUATION

The study will report tokenizer-level compactness, spelling-variant collision rates, language-model loss or perplexity, and downstream sentiment performance. The primary robustness comparison will examine the relative degradation between clean and spelling-perturbed evaluation conditions. No conclusion is drawn in advance about which representation will perform better; the experimental results will determine whether phonetic tokenization provides an advantage under the proposed conditions.

V. REFERENCES

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